Titanic Survival Prediction Using Machine Learning

1. Introduction

Titanic Survival Prediction is a machine learning classification project that predicts whether a passenger survived the Titanic disaster based on passenger information such as age, gender, passenger class, fare, family size, and embarkation port. This project demonstrates a complete machine learning workflow, including data preprocessing, exploratory data analysis (EDA), feature engineering, model training, and model evaluation.

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2. Objective

The main objectives of this project are:

- To analyze the Titanic dataset.
- To clean and preprocess the data.
- To perform Exploratory Data Analysis (EDA).
- To prepare the dataset for machine learning.
- To train multiple machine learning classification models.
- To compare the performance of different models.
- To identify the best-performing model for predicting passenger survival.

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3. Dataset Description

The dataset used in this project is the **Titanic Dataset**.

Dataset Information

- **Total Records:** 891
- **Total Features:** 12
- **Target Variable:** Survived

The dataset contains passenger details such as age, gender, passenger class, fare, family information, ticket details, cabin information, and embarkation port.

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4. Data Preprocessing

The following preprocessing steps were performed:

- Loaded the Titanic dataset.
- Checked dataset dimensions and data types.
- Identified missing values.
- Filled missing numerical values.
- Filled missing categorical values.
- Removed unnecessary columns.
- Encoded categorical variables.
- Performed feature scaling.
- Checked duplicate records.
- Split the dataset into training and testing sets.

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5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the dataset and identify patterns affecting passenger survival.

The following analyses were carried out:

- Distribution of passenger survival.
- Survival analysis based on gender.
- Passenger class analysis.
- Age distribution analysis.
- Fare distribution analysis.
- Correlation analysis between numerical features.

These analyses helped in understanding the important features influencing passenger survival and improved the machine learning model development process.

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6. Feature Engineering

Relevant features were selected and prepared for model training. Categorical variables were transformed into numerical format, and the dataset was processed to improve model performance.

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7. Data Encoding

Categorical variables were encoded into numerical values so that machine learning algorithms could process the data efficiently.

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8. Machine Learning Models

The following classification algorithms were trained and evaluated:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

The dataset was divided into training and testing sets using an 80:20 ratio.

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9. Model Evaluation

The models were evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1 Score

Model Performance

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.7989	0.7879	0.7027	0.7429
Decision Tree Classifier	0.8101	0.8030	0.7162	0.7571
Random Forest Classifier	0.8156	0.7887	0.7568	0.7724

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10. Results

Among the three classification algorithms, the Random Forest Classifier achieved the best overall performance.

Best Model

Random Forest Classifier

Performance

- Accuracy: 81.56%
- Precision: 78.87%
- Recall: 75.68%
- F1 Score: 77.24%

The Random Forest Classifier achieved the highest Accuracy, Recall, and F1 Score, making it the most suitable model for predicting passenger survival in this project.

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11. Conclusion

This project successfully developed a machine learning model for predicting Titanic passenger survival using the Titanic dataset. The project followed a complete machine learning workflow, including data preprocessing, exploratory data analysis, feature engineering, model training, and model evaluation.

Three classification models Logistic Regression, Decision Tree Classifier, and Random Forest Classifier were trained and compared. Based on the evaluation metrics, the Random Forest Classifier achieved the best overall performance with an Accuracy of 81.56%, Precision of 78.87%, Recall of 75.68%, and an F1 Score of 77.24%.

This project demonstrates practical skills in data preprocessing, exploratory data analysis, feature engineering, machine learning model development, and performance evaluation using Python and Scikit-learn.

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12. Future Scope

The project can be further improved by:

- Applying hyperparameter tuning to improve model performance.
- Testing advanced classification algorithms such as XGBoost, LightGBM, and Support Vector Machine (SVM).
- Deploying the model using Flask or Streamlit.
- Developing an interactive web application for real-time Titanic survival prediction.

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13. References

1. Kaggle – Titanic: Machine Learning from Disaster Dataset
2. Scikit-learn Documentation
3. Pandas Documentation
4. NumPy Documentation
5. Matplotlib Documentation
6. Seaborn Documentation